



Fig. 5: Suggested installation for the shadow alarm

tor. Connect the 9V power supply at J1 connector. The Arduino board will receive power at J2 connector.

When power to the circuit is switched on by closing switch S1, the power LED in the Arduino board will glow.

Now download the latest Arduino

software from <http://www.arduino.cc/en/main/software> and install it in your PC. Compile the sketch (Shadow_Alarm.pde) listed at the end of this article. Upload the code into ATmega328 microcontroller on the Arduino board.

Testing of the circuit is simple. Point LDR1 towards a light source such as bulb or tubelight. Using your hand, block the light falling on LDR1. You should hear the alarm sound from piezobuzzer PZ1. You can vary VR1 to adjust the sensitivity

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

of LDR1 sensor. When you remove your hand, the buzzer should stop and LED1 glow. This completes the testing.

In case of any problem in the circuit, you can refer to Test Points table. The table shows voltage status at various test points in the circuit.

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SHADOW_ALARM.PDE

```
//Shadow alarm
int rlyPin = 12; // Relay Output Pin
int sensPin = 2; // Sensor Input Pin
int ledPin = 8; // Reminder LED Output Pin
int pzSpeaker = 10; //Piezo-speaker Output Pin
int val = 0; // variable for reading the Input Pin status
void setup() {
  pinMode(rlyPin, OUTPUT); // Set Relay as output
  pinMode(sensPin, INPUT); // Set Shadow Sensor as input
  pinMode(ledPin, OUTPUT); // Set LED as output
  pinMode(pzSpeaker, OUTPUT); // Set Piezo-Speaker as output
}
void loop() {
  val = digitalRead(sensPin); // read input value
  if (val == HIGH) { // check if the input is HIGH
    digitalWrite(rlyPin, HIGH); // turn Relay ON
    digitalWrite(ledPin, HIGH); // turn LED ON
    playTone(500, 600);
    delay(100);
    playTone(500, 800);
    delay(100);
  }
  else {
    digitalWrite(rlyPin, LOW); // turn Relay OFF
    playTone(0, 0);
    delay(300);
  }
  // duration in mSecs, frequency in hertz
  void playTone(long duration, int freq) {
    duration *= 1000;
    int period = (1.0 / freq) * 1000000;
    long elapsed_time = 0;
    while (elapsed_time < duration) {
      digitalWrite(pzSpeaker, HIGH);
      delayMicroseconds(period / 2);
      digitalWrite(pzSpeaker, LOW);
      delayMicroseconds(period / 2);
      elapsed_time += (period);
    }
  }
}
```

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